

Comparative PM_{2.5} Time Series Forecasting with Extreme-Event and Statistical Evaluation

Kiet Gia Tran^[0009–0008–7020–0944], Thu Le^[0009–0008–3480–8311], Cuong Phu Le^[0009–0007–0536–2843], Tien Manh Tran Chau^[0009–0003–3502–4838], and Nghia Trong Nguyen Ho^[0009–0007–3460–8814]

FPT University, Ho Chi Minh City, Viet Nam
trankietworkmail@gmail.com, thulvm@fpt.edu.vn, lpcuong2006@gmail.com,
chautranmanhtien@gmail.com, trongnghiaho104@gmail.com

Abstract. Accurate short-term forecasting of fine particulate matter ($PM_{2.5}$) is essential for air quality management and public health protection, yet systematic comparative evaluations across multiple forecast horizons with statistical significance testing remain limited.

This work evaluated five models—Linear Regression, DLinear, LSTM, PatchTST, and XGBoost—for hourly $PM_{2.5}$ prediction at Station 33 (Randwich) in Sydney across horizons $H \in \{1, 3, 6, 12, 24\}$ hours, using MAE, RMSE, R^2 , Diebold–Mariano testing, and extreme-event metrics (Peak MAE, Spike Recall) for high-pollution episodes (top 10%). Results reveal a horizon-dependent performance transition: LSTM dominates at H_1 ($R^2 = 0.733$, MAE = $1.725 \mu g/m^3$, Peak MAE = $3.38 \mu g/m^3$), while XGBoost leads in Spike Recall; at H_{24} , PatchTST and LSTM become statistically interchangeable ($p = 0.349$) as linear models collapse ($R^2 < 0$). These findings underscore a trade-off between accuracy and extreme-event detection, highlighting the need for extreme-aware evaluation in operational air quality forecasting.

Keywords: $PM_{2.5}$ · Time Series Forecasting · Multi-horizon Prediction · Machine Learning · Diebold–Mariano Test · Extreme Event Detection · Air Quality Forecasting

1 Introduction

Accurate forecasting of fine particulate matter ($PM_{2.5}$) is central to air quality management and public health protection. Urban $PM_{2.5}$ dynamics arise from complex interactions between traffic emissions, meteorological conditions, and regional transport, rendering the task inherently nonlinear and non-stationary.

Recent data-driven approaches have advanced single-horizon prediction, yet systematic comparative evaluations remain limited. Existing studies evaluate models at fixed short horizons without characterizing accuracy decline over longer lead times [17, 18], omit formal statistical significance testing, and rely solely on global accuracy metrics that obscure behavior during high-pollution episodes [9, 16]. This work addresses these gaps through a systematic comparison

of five models—Linear Regression, DLinear, LSTM, PatchTST, and XGBoost—on an hourly $PM_{2.5}$ dataset from Sydney (2021–2025) across horizons $H \in \{1, 3, 6, 12, 24\}$ hours.

Contributions

Multi-horizon diagnostic pipeline. Residual PDF, ACF analysis, and extreme-event evaluation ($PM_{2.5} > P90$) enable structured diagnosis of accuracy deterioration across forecast lead times.

Statistical significance validation. Diebold–Mariano testing formally validates whether metric differences reflect genuine predictive superiority across H_1 and H_{24} .

Practitioner-oriented guidance. Empirically grounded model-selection guidance for operational $PM_{2.5}$ forecasting across early-warning and longer-range planning horizons.

The remainder covers related work (Sec. 2), methodology (Sec. 3), experimental setup (Sec. 4), results (Sec. 5), and conclusion (Sec. 6).

2 Related Work

2.1 $PM_{2.5}$ Forecasting with Machine Learning

Early approaches relied on ARIMA and linear regression, which struggle with nonlinear atmospheric dynamics [10, 11]. LSTM networks substantially improved accuracy by modeling short-term sequential dependencies [9, 17], with spatial context from neighboring stations providing complementary signals [8, 18]. XGBoost [7] shows competitive performance with meteorological feature engineering, and spatiotemporal fusion networks extend prediction to city-wide scales [14, 16, 15]. Despite these advances, most studies evaluate a single short horizon without characterizing skill decline or applying formal statistical testing.

2.2 Transformer and Linear Architectures for Time-Series Forecasting

Transformer-based models—Informer [6], Autoformer [3], and PatchTST [2]—replace sequential processing with attention mechanisms [5] for superior long-horizon forecasting. PatchTST has been applied to air quality forecasting [19] and environmental transfer learning [20]. Zeng et al. [1] showed that simple linear mappings (DLinear) can outperform complex Transformers on data with strong trend components. Our work bridges both streams within a unified multi-horizon evaluation under a challenging right-skewed $PM_{2.5}$ distribution (mean = $6.14 \mu g/m^3$, max = $556.29 \mu g/m^3$) with explicit focus on extreme pollution episodes.

3 Methodology

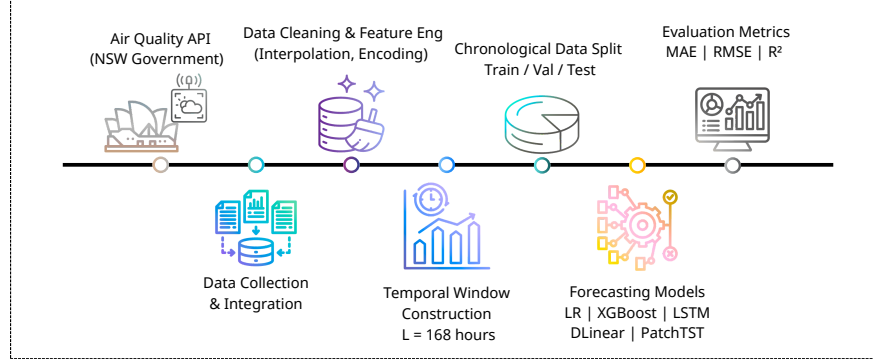


Fig. 1. Overall $PM_{2.5}$ forecasting pipeline.

3.1 Problem Formulation

Let $y_t \in \mathbb{R}$ denote the $PM_{2.5}$ concentration at Station 33 at time t , measured in $\mu g/m^3$. Given a look-back window of $L = 168$ hours (one week), the input matrix $X_{t-L+1:t} \in \mathbb{R}^{L \times F}$ comprises air-quality observations ($PM_{2.5}$, PM_{10} , NO_2 , O_3) from Stations 33, 39, and 206, meteorological variables (wind speed, wind direction, rainfall, temperature), and cyclical temporal encodings via sine-cosine transformation. A direct multi-step strategy is adopted:

$$\hat{y}_{t+H} = f_H(X_{t-L+1:t}), \quad H \in \{1, 3, 6, 12, 24\},$$

where f_H is the learned mapping for horizon H . All models are benchmarked against a Persistence baseline $\hat{y}_{t+H} = y_t$.

3.2 Forecasting Models

Linear baselines. Linear Regression captures linear relationships between the flattened input window and the forecast target. DLinear [1] decomposes the input into trend and remainder via a moving average filter, applying independent linear projections to each.

Nonlinear models. XGBoost [7] models complex nonlinear interactions through gradient-boosted decision trees with the $L \times F$ window flattened into a single feature vector. LSTM [4] captures sequential dependencies via gated recurrent connections, motivated by the strong lag-1 autocorrelation and diurnal periodicity of the data (Section 4.3).

Transformer-based model. PatchTST [2] applies self-attention across non-overlapping patch tokens with channel independence enforced, hypothesized to offer stability advantages at H_{24} .

3.3 Evaluation Protocol

Standard accuracy metrics. MAE [12] measures average prediction error on the original $\mu\text{g}/\text{m}^3$ scale. RMSE [13] penalizes large deviations more heavily. R^2 quantifies variance explained, with $R^2 < 0$ indicating the model is outperformed by the mean predictor.

Residual diagnostics. Residual PDFs characterize bias (μ) and dispersion (σ) at H_1 and H_{24} ; ACFs detect remaining temporal structure.

Extreme-event evaluation. Peak MAE measures error at time steps where $PM_{2.5}$ exceeds the 90th percentile. Spike Recall measures the fraction of such episodes correctly identified via threshold-based classification.

Statistical significance testing. The Diebold–Mariano (DM) test [12] under absolute-error loss assesses whether accuracy differences between model pairs are statistically significant.

4 Experimental Setup

4.1 Data Collection and Study Area

The dataset was obtained from the NSW Department of Planning and Environment through the official Air Quality API¹. It contains hourly observations (2021–2025) in Sydney East, Australia. Station 33 (Randwick) is the forecasting target; Stations 39 (Rozelle) and 206 (Earlwood) serve as exogenous predictors (Fig. 2).



Fig. 2. Geographical locations of monitoring stations. Red: target 33 (Randwick); Blue: predictors 39 (Rozelle) and 206 (Earlwood).

¹ <https://www.airquality.nsw.gov.au/air-quality-data-services/air-quality-api>

4.2 Data Preprocessing, Feature Engineering, and Partitioning

The raw dataset (1,136,304 records) was reshaped from long to wide format. After removing 95 duplicates, 101,699 missing values were addressed: rainfall gaps zero-filled, remaining gaps linearly interpolated (max 6 hours) then forward-filled. The final dataset comprises 37,735 hourly observations. Pollutant values were clipped to non-negative values.

Cyclical features were encoded via sine–cosine transformation: $Wind_{\sin} = \sin(\theta)$, $Wind_{\cos} = \cos(\theta)$, and analogously for hour-of-day, day-of-week, and month-of-year. All features were standardized using `StandardScaler` fitted on the training partition only, preventing information leakage. The 37,735 effective samples are split chronologically into 70% training, 15% validation, and 15% test.

4.3 Exploratory Data Analysis

Summary statistics (Table 1) reveal a significant right-skew: mean $PM_{2.5}$ is $6.14 \mu g/m^3$ with extremes reaching $556.29 \mu g/m^3$, and an ADF test (-17.86 , $p < 0.001$) confirms stationarity. $PM_{2.5}$ shows strong positive associations with PM_{10} and NO_2 and negative correlations with wind speed and ozone. ACF analysis (Fig. 3) reveals 24-hour periodicity and a dominant PACF spike at lag 1, motivating the inclusion of LSTM.

Table 1. Statistical summary and top Pearson correlations (r) for $PM_{2.5}$ at Station 33.

Descriptive Statistics			Top Correlations (r)		
Metric	y_t	y_{t+24}	Variable	Station	Corr.
Mean	6.1396	6.1380	PM_{10}	33	+0.581
Std	7.0444	7.0424	PM_{10}	206	+0.581
Median	5.129	5.129	NO_2	39	+0.199
P99	23.780	23.775	WSP	33	−0.103
Max	556.29	556.29	$Ozone$	33	−0.099

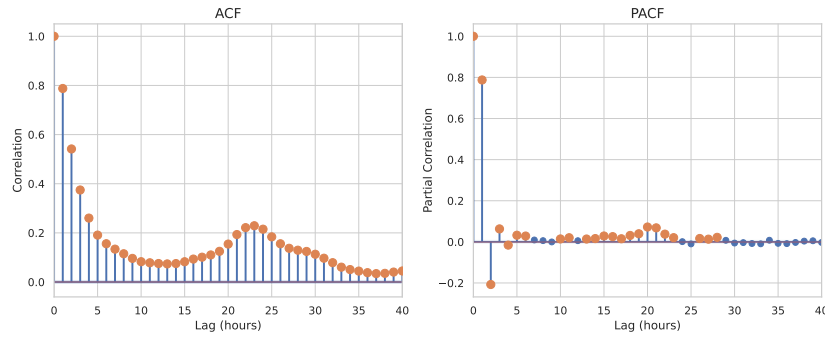


Fig. 3. ACF and PACF of $PM_{2.5}$ at Station 33, highlighting strong short-term dependence (lag 1) and 24-hour diurnal periodicity.

Seasonal analysis (Table 2) reveals that Spring exhibits the highest variance (std = $10.64 \mu\text{g}/\text{m}^3$, max = $556.29 \mu\text{g}/\text{m}^3$), driven by bushfire smoke events, motivating cyclical temporal encodings. Monthly medians remain stable with slight elevations in September and December (Fig. 4).

Table 2. Seasonal statistical summary of $PM_{2.5}$ concentrations at Station 33 ($\mu\text{g}/\text{m}^3$).

Season	Mean	Std	Min	Max	Season	Mean	Std	Min	Max
Summer	6.52	5.26	0.00	58.53	Winter	5.93	5.44	0.00	77.54
Autumn	5.85	4.71	0.00	66.93	Spring	6.29	10.64	0.00	556.29

Monthly statistics show that median $PM_{2.5}$ levels remain relatively stable throughout the year. Slight increases during September and December suggest a weak seasonal influence.

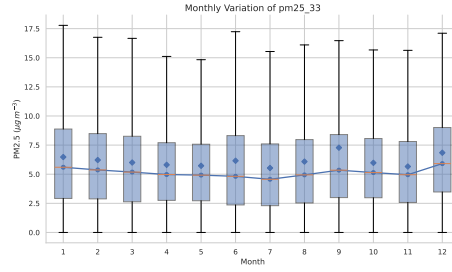


Fig. 4. Monthly distribution of $PM_{2.5}$ concentrations.

The heatmap reveals a pronounced diurnal pattern, with peak concentrations typically occurring during early morning hours (05:00–07:00), while weekly variation appears comparatively weak.

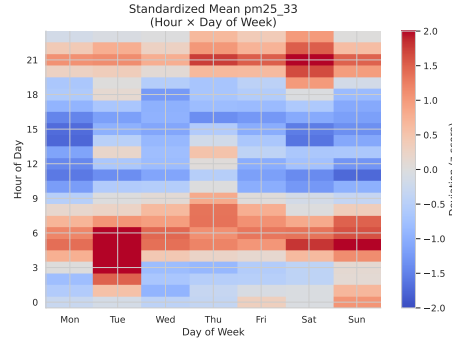


Fig. 5. Diurnal and weekly patterns of normalized $PM_{2.5}$ concentrations.

5 Results and Discussion

This section examines the five forecasting models across four analytical dimensions: (i) standard accuracy metrics across all horizons (Section 5.1), (ii) residual distribution and autocorrelation structure (Sections 5.2–5.3), (iii) robustness under high-pollution events (Section 5.4), and (iv) formal statistical significance testing via the Diebold–Mariano test (Sections 5.5–5.6).

5.1 Multi-Horizon Accuracy

Performance declines monotonically with horizon, confirming progressive decay of temporal dependencies in $PM_{2.5}$ dynamics. LSTM dominates H1–H12, peaking at $R^2 = 0.733$ at H1 and remaining the most resilient at H12 ($R^2 = 0.155$), while PatchTST overtakes at H24 ($R^2 = 0.028$). Linear models degrade substantially beyond H6 ($R^2 < -0.2$ at H24).

Table 3. Test-set performance across models and forecast horizons. MAE and RMSE are in $\mu g/m^3$; R^2 is dimensionless. Asterisk (*) denotes the best value per metric and horizon.

Model	Metric	H1	H3	H6	H12	H24
Linear	MAE	2.099	2.836	3.205	3.538	3.656
	RMSE	2.781	3.906	4.387	4.906	5.091
	R^2	0.638	0.285	0.099	-0.127	-0.213
DLinear	MAE	2.049	2.865	3.242	3.550	3.665
	RMSE	2.728	3.930	4.418	4.921	5.088
	R^2	0.651	0.277	0.086	-0.134	-0.211
LSTM	MAE	1.725*	2.326*	2.812*	2.902*	3.157
	RMSE	2.385*	3.387*	3.996*	4.249*	4.619
	R^2	0.733*	0.463*	0.253*	0.155*	0.002
PatchTST	MAE	2.316	2.619	2.915	3.005	3.119*
	RMSE	3.311	3.699	4.137	4.410	4.558*
	R^2	0.486	0.359	0.199	0.089	0.028*
XGBoost	MAE	1.819	2.571	2.847	3.161	3.284
	RMSE	2.750	4.063	4.160	4.584	4.644
	R^2	0.646	0.227	0.190	0.016	-0.009

5.2 Residual Distribution Analysis

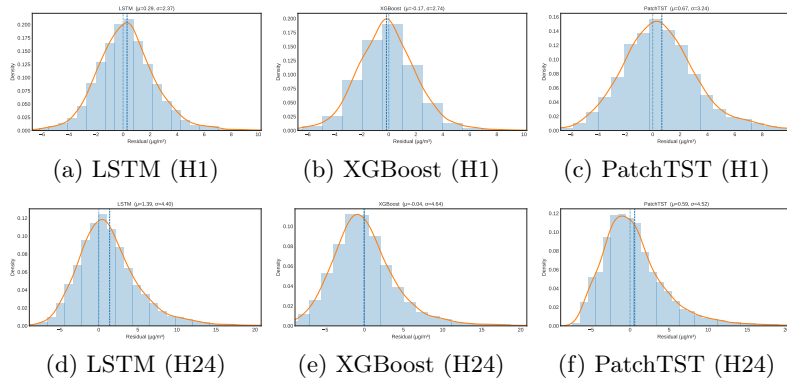


Fig. 6. Residual distributions at H1 and H24. LSTM (H1: $\mu=0.29, \sigma=2.37$; H24: $\mu=1.39, \sigma=4.40$), XGBoost (H1: $\mu=-0.17, \sigma=2.74$; H24: $\mu=-0.04, \sigma=4.64$), PatchTST (H1: $\mu=0.67, \sigma=3.24$; H24: $\mu=0.59, \sigma=4.52$).

At H1, LSTM achieves the narrowest error distribution ($\sigma = 2.37 \mu g/m^3$) while XGBoost exhibits the lowest bias ($\mu = -0.17 \mu g/m^3$). DLinear improves over Linear through decomposition, reducing both bias (0.29 vs 0.42) and dispersion (2.71 vs 2.75). PatchTST shows higher initial dispersion ($\sigma = 3.24$), trading short-term precision for longer-horizon stability. At H24, all models experience variance inflation: LSTM remains most precise ($\sigma = 4.40 \mu g/m^3$) but develops systematic underestimation bias ($\mu = 1.39 \mu g/m^3$), likely due to regression-to-the-mean at longer horizons. XGBoost maintains the lowest bias ($\mu = -0.04 \mu g/m^3$), while Linear and DLinear converge to high dispersions ($\sigma \approx 5.00 \mu g/m^3$), confirming linear decomposition cannot sustain predictive structure beyond short forecast windows.

5.3 Residual Autocorrelation Analysis

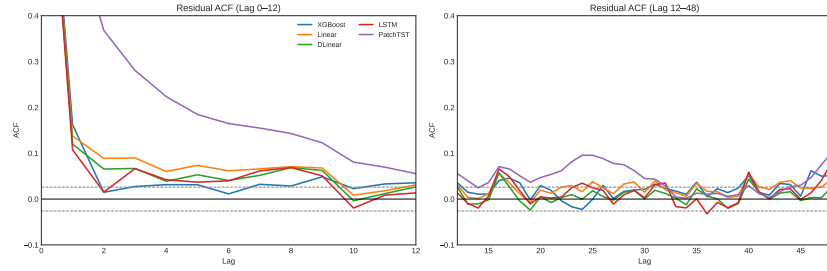


Fig. 7. Residual ACF at H1. Dashed lines denote the 95% confidence interval; residuals close to white noise indicate well-specified models.

At H1 (Fig. 7), LSTM and XGBoost exhibit the fastest autocorrelation decay toward zero, with residuals approximating white noise and confirming adequate capture of short-term temporal structure. PatchTST shows slightly stronger short-lag persistence, and most models remain within the 95% confidence bounds beyond lag 3.

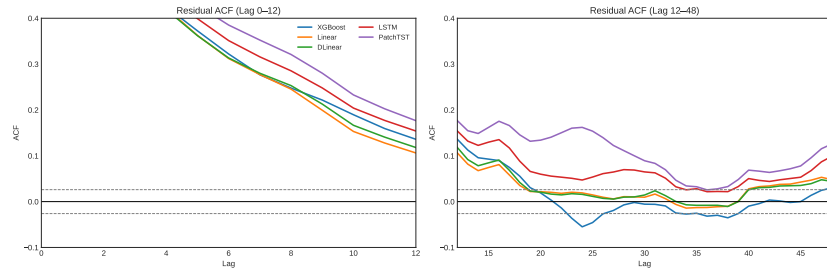


Fig. 8. Residual ACF at H24. Increased persistence indicates unresolved temporal structure at longer forecast lead times.

At H24 (Fig. 8), residual autocorrelation becomes more pronounced across all models. Linear and DLinear exhibit notable short-lag correlations indicating

substantial uncaptured temporal structure, while LSTM and XGBoost remain comparatively stable. Persistent correlations at longer lags confirm that $PM_{2.5}$ dynamics become increasingly difficult to model as the horizon extends, likely reflecting synoptic-scale meteorological variability beyond the 168-hour input window.

5.4 Performance Under High-Pollution Events

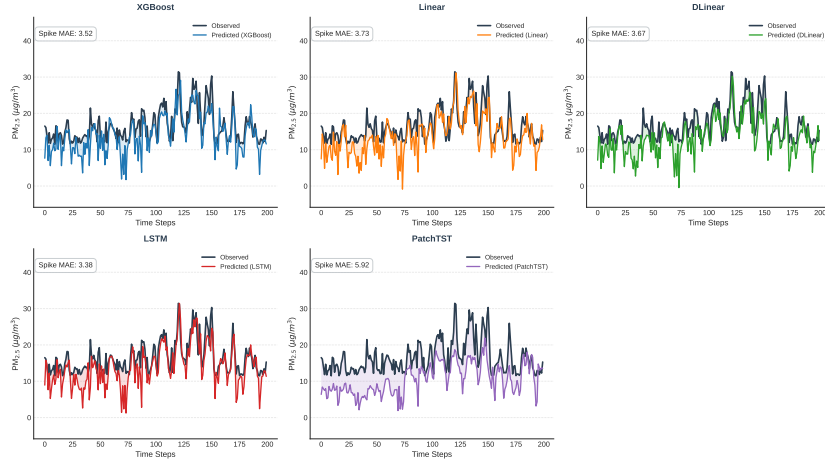


Fig. 9. Model predictions during extreme $PM_{2.5}$ events (top 10%) at H1. LSTM tracks magnitude most closely; XGBoost identifies more spike occurrences.

Table 4. Extreme-event forecasting performance ($PM_{2.5} > P90$) at H1.

Model	Peak MAE ↓	Spike Recall ↑
LSTM	3.38*	0.63
XGBoost	3.52	0.67*
DLinear	3.67	0.60
Linear	3.73	0.59
PatchTST	5.92	0.36

Table 4 and Fig. 9 present model performance at time steps where $PM_{2.5}$ exceeds $P90$, targeting high-risk scenarios most critical for public health intervention. LSTM achieves the lowest Peak MAE ($3.38 \mu g/m^3$), most accurately estimating episode intensity, while XGBoost attains the highest Spike Recall (0.67), identifying a greater fraction of extreme events. This trade-off has direct operational implications: applications requiring precise concentration forecasts favor LSTM, whereas early-warning systems prioritizing detection may benefit from XGBoost. PatchTST performs substantially worse under extreme conditions (Peak MAE = $5.92 \mu g/m^3$, Recall = 0.36), indicating limited responsiveness to abrupt surges despite its competitive performance at H24—suggesting its

long-range attention mechanism does not confer advantages for short-term spike detection. Crucially, these findings reveal that global metrics (MAE, RMSE, R^2) are insufficient alone: a model with superior global performance may not dominate on extreme-event detection, and vice versa.

5.5 Statistical Significance Test at H1

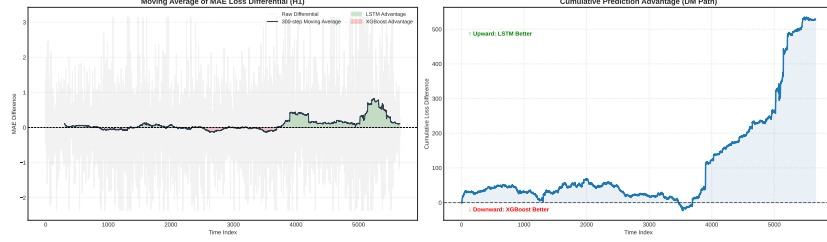


Fig. 10. Diebold–Mariano analysis between LSTM and XGBoost at H1. A persistent upward cumulative advantage path confirms LSTM’s superiority throughout the evaluation period.

The DM test under absolute-error loss compares LSTM and XGBoost—the two strongest-performing models at H1. LSTM outperforms XGBoost in 51.63% of time steps, with a persistent upward cumulative advantage path (Fig. 10), confirming that LSTM’s advantage is consistently maintained rather than concentrated in specific periods. The mean loss differential is $0.0933 \mu\text{g}/\text{m}^3$ and the DM statistic is 4.69 ($p = 2.67 \times 10^{-6}$), rejecting equal predictive accuracy at the 0.1% significance level.

5.6 Statistical Significance Test at H24

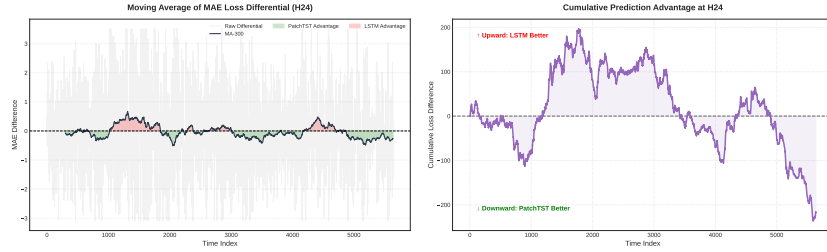


Fig. 11. Diebold–Mariano comparison between PatchTST and LSTM at H24. The fluctuating cumulative path and $p = 0.349$ confirm statistically comparable performance.

At H24, the DM test between PatchTST and LSTM yields a statistic of -0.937 ($p = 0.349$), indicating no statistically significant difference in predictive accuracy. PatchTST achieves lower errors in 51.47% of time steps, but the cumulative advantage path fluctuates without a consistent trend (Fig. 11), and the mean loss

differential of $-0.0385 \mu\text{g}/\text{m}^3$ is negligible in practice. Accordingly, PatchTST and LSTM are statistically interchangeable for long-horizon $PM_{2.5}$ forecasting under this experimental setting.

5.7 Summary and Model-Selection Guidance

Across all horizons, predictive accuracy degrades monotonically as temporal dependencies weaken. LSTM dominates H1–H12, offering the lowest error and strong robustness under high-pollution conditions, though XGBoost achieves higher Spike Recall at H1 despite lower overall accuracy. At H24, PatchTST and LSTM are statistically interchangeable ($p = 0.349$), while linear models collapse below the persistence baseline.

For rapid early-warning (H1), LSTM is recommended for lowest overall error; XGBoost is preferable when extreme-event detection is the primary objective. For H3–H6, LSTM remains preferred, maintaining $R^2 > 0.25$ through H6. For H12–H24, PatchTST and LSTM are statistically comparable and either is appropriate, while linear models should be avoided. Across all horizons, extreme-event metrics (Peak MAE, Spike Recall) should be reported alongside MAE/RMSE/ R^2 to enable informed operational decisions.

6 Conclusion

This work assessed five forecasting models for hourly $PM_{2.5}$ prediction across $H \in \{1, 3, 6, 12, 24\}$ hours within a single comparative protocol integrating standard accuracy metrics, residual diagnostics, extreme-event evaluation, and Diebold–Mariano statistical testing.

Results expose a clear horizon-dependent performance transition. At H1, LSTM achieves the strongest overall accuracy and lowest Peak MAE, demonstrating the suitability of recurrent architectures for short-term forecasting. However, XGBoost attains higher Spike Recall, revealing an important trade-off: models optimized for global error minimization may underperform in detecting critical high-pollution episodes. At H24, PatchTST and LSTM exhibit statistically comparable performance ($p = 0.349$), while linear models degrade substantially ($R^2 < 0$), underscoring the advantage of nonlinear architectures under increasing temporal uncertainty.

Residual analysis uncovers growing variance and persistent autocorrelation at longer horizons, highlighting the limitations of single-station fixed-window forecasting in capturing synoptic-scale atmospheric processes, and calling for extreme-aware assessment criteria beyond conventional MAE/RMSE reporting. The evaluation is confined to a single station (Randwick), and generalizability to other urban environments remains to be verified. Extreme-event evaluation is restricted to H1; whether the accuracy–detection trade-off persists at longer horizons is an open question. Future work should extend this approach to multi-city datasets, explore ensemble and hybrid architectures targeting improved extreme-event responsiveness, and investigate spatiotemporal models leveraging inter-station dependencies more explicitly.

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